

PROBABILITY THEORY

Session 2

RANDOM VARIABLES

Topics:

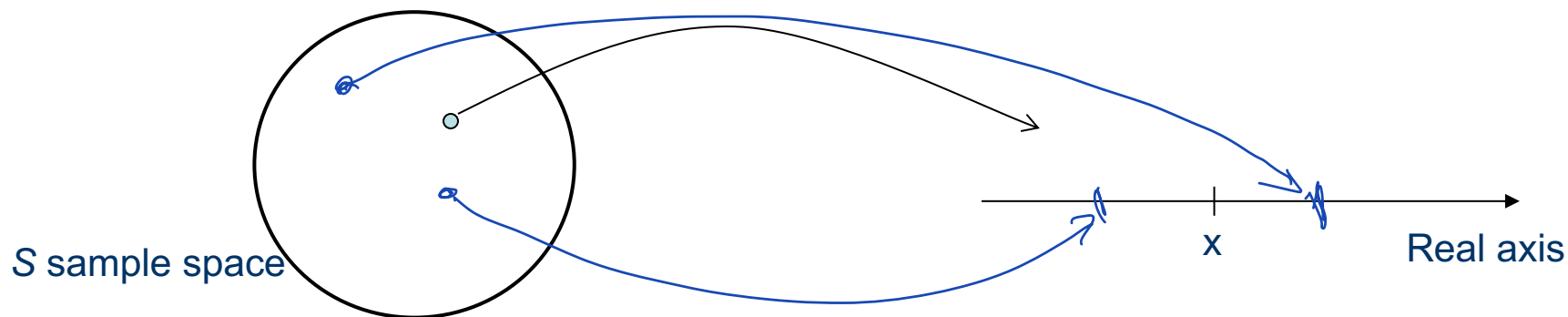
Discrete and continuous random variables.
Probability mass function and probability density function.
Cumulative distribution function.
Jointly distributed random variables.
Independent random variables.
Expectation. Variance. Covariance.

What should we learn today?

- What is a random variable?
- What types of random variables exist – how are they different and similar?
- How to work with distributions:
 - Cumulative density function
 - Probability mass function
 - Probability density function
- How to handle jointly distributed multiple random variables?
- How to define and work with expectation, variance and covariance

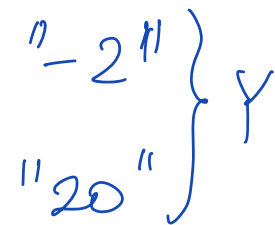
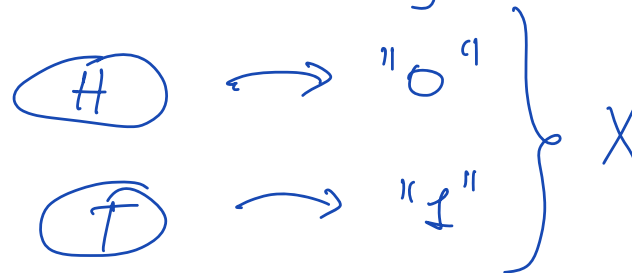
Random variable

- **Definition.** A r.v. X is a function that assigns a real number to each outcome of a random experiment.



Examples

- A r.v. X is a lifetime of a system component. } cont. r.v.
- Time until a next job arrives at a server
- Number of trees on a square meter } discrete r.v.
- Coin flipping



Example

- A coin is flipped 3 times. A r.v. X is a number of heads in 3 trials.

$$X \in \{0, 1, 2, 3\}$$

$$P(X=3) = \frac{\text{\# of out. when 3 Heads}}{\text{total \# of outcomes}} = \frac{1}{8}$$

$$\begin{array}{c} \text{H/T} \\ \text{---} \\ \text{O} \end{array} \cdot \begin{array}{c} \text{H/T} \\ \text{---} \\ \text{O} \end{array} \cdot \begin{array}{c} \text{H/T} \\ \text{---} \\ \text{O} \end{array} = 2^3 = 8$$

$$\begin{array}{c} \text{H} \\ \text{---} \\ \text{O} \end{array} \quad \begin{array}{c} \text{H} \\ \text{---} \\ \text{O} \end{array} \quad \begin{array}{c} \text{H} \\ \text{---} \\ \text{O} \end{array}$$

$$P(X=5) = 0$$

$$P(X=1.5) = 0$$

$$P(X=2) = \frac{3}{8}$$

$$\left. \begin{array}{ccc} \begin{array}{c} \text{H} \\ \text{---} \\ \text{O} \end{array} & \begin{array}{c} \text{T} \\ \text{---} \\ \text{O} \end{array} & \begin{array}{c} \text{H} \\ \text{---} \\ \text{O} \end{array} \\ \text{H} & \text{H} & \text{T} \\ \text{T} & \text{H} & \text{H} \end{array} \right\} 3$$

$$\binom{3}{2} = \frac{3!}{2!(3-2)!} = \frac{3!}{2! \cdot 1!} = 3$$

Example: coin flipping

Possible outcomes:

HHH	3
HHT	2
HTH	2
THH	2
HTT	1
THT	1
TTH	1
TTT	0

$$P(X=3) = \frac{1}{8}$$

$$P(X=2) = \frac{3}{8}$$

$$P(X=1) = \frac{3}{8}$$

$$P(X=0) = \frac{1}{8}$$

prob. mass function

$$P(X \leq 2) =$$
$$= P(\underbrace{X=0} \cup \underbrace{X=1} \cup \underbrace{X=2}) =$$

mutually excl.

$$= P(X=0) + P(X=1) + P(X=2) =$$
$$= \frac{1}{8} + \frac{3}{8} + \frac{3}{8} = \frac{7}{8}$$

$$P(X > 1)$$

Example: coin flipping

Possible outcomes:

HHH	3
HHT	2
HTH	2
THH	2
HTT	1
THT	1
TTH	1
TTT	0

$$P(X=3) = \frac{1}{8}$$

$$P(X=2) = \frac{3}{8}$$

$$P(X=1) = \frac{3}{8}$$

$$P(X=0) = \frac{1}{8}$$

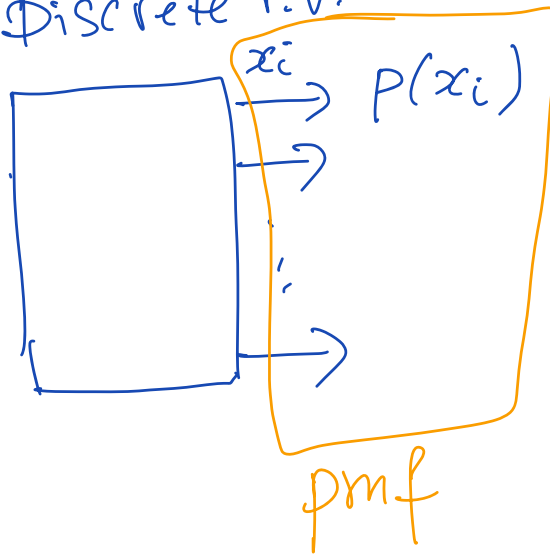
$$\begin{aligned} \text{Total number of outcomes} &= \\ &= 2^3 = 8 \end{aligned}$$

$$\begin{aligned} \text{\# of outcomes when \# of H is } i \\ \binom{3}{i} = \frac{3!}{(3-i)!i!} = \begin{cases} 1, & i=3 \\ 3, & i=2 \\ 3, & i=1 \\ 1, & i=0 \end{cases} \end{aligned}$$

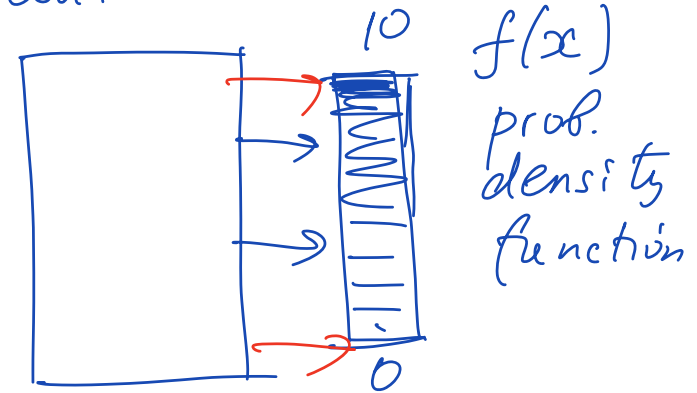
$$P(X \leq 2) = P(X=0) + P(X=1) + P(X=2) = \frac{7}{8}$$

How to describe behaviour of a r.v.?

Discrete r.v.



Cont. r.v.



Discrete r.v.

- **Definition.** A r.v. whose set of possible values is a sequence is said to be **discrete**.
- For a discrete r.v. we define **probability mass function**

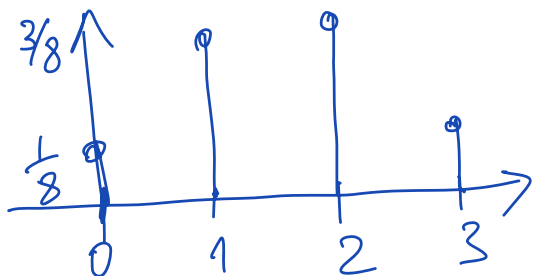
$$p(a) = P(X = a)$$

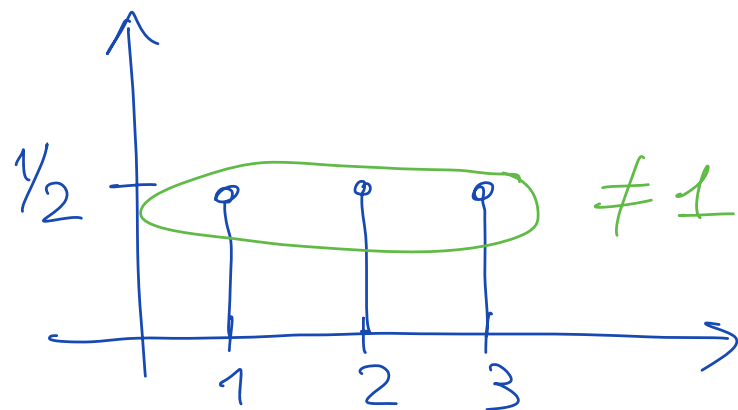
$$p(x_i) > 0, \quad i = 1, 2, \dots$$

$$p(x) = 0, \quad \text{otherwise}$$

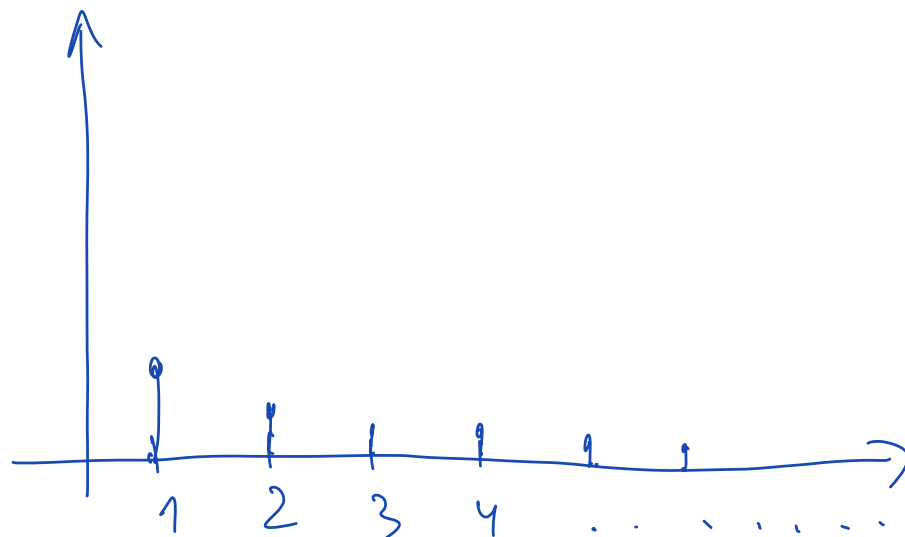
$$\sum_{i=1}^{\infty} p(x_i) = 1$$

Ex. coin flipping



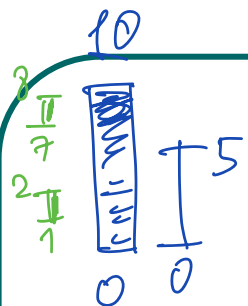


$\Rightarrow$ not a pmf



$$\sum_{i=1}^{\infty} p(x_i) = 1$$

Example coin flipping



$$\int_0^5 f(x) dx = P(X \in [0, 5])$$

Continuous r.v.

- **Definition.** X is a **continuous r.v.** if there exists a nonnegative function $f(x)$, defined for all x having the property that for any set B of real numbers

$$P(X \in B) = \int_B f(x) dx$$

- Function $f(x)$ is called the **probability density function** of X

- Additionally, $f(x)$ should satisfy

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

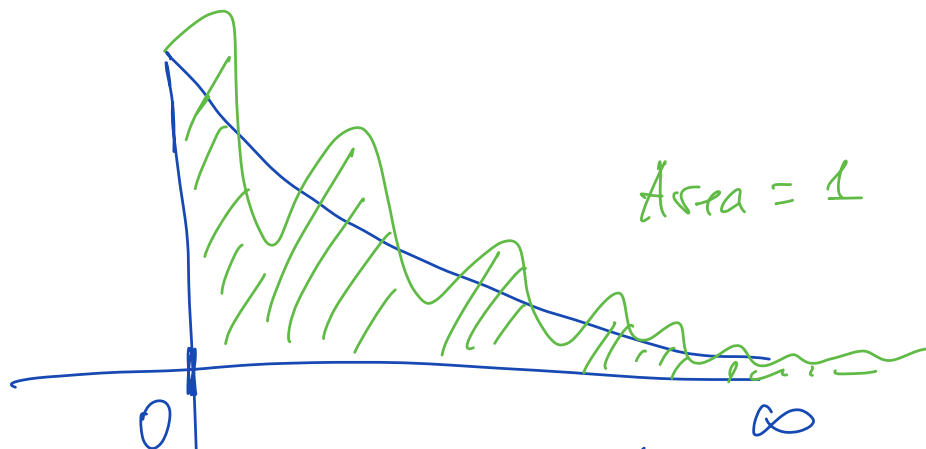
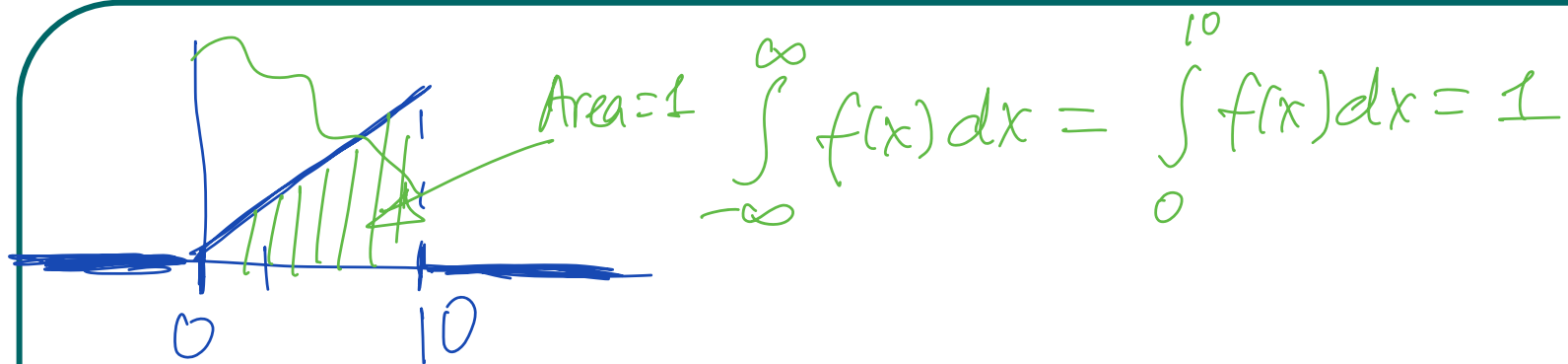
- Different probability statements can be expressed using pdf

- If we let $a=b$, then

$$P(a \leq X \leq b) = \int_a^b f(x) dx$$

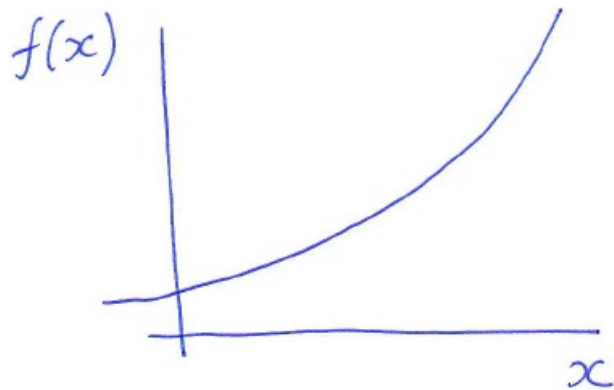
$$P(X = a) = \int_a^a f(x) dx = 0$$

- Probability that a continuous r.v. will assume any particular value is zero

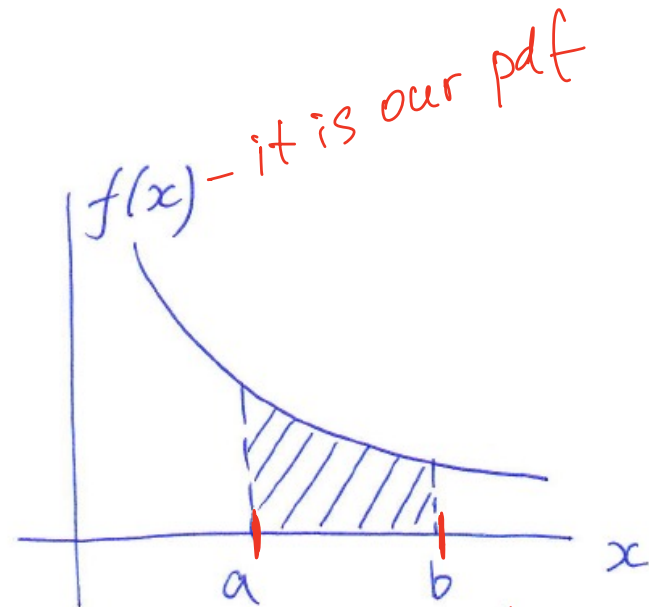


$f(x) = e^x \rightarrow$ this can not be a pdf

Continuous r.v.

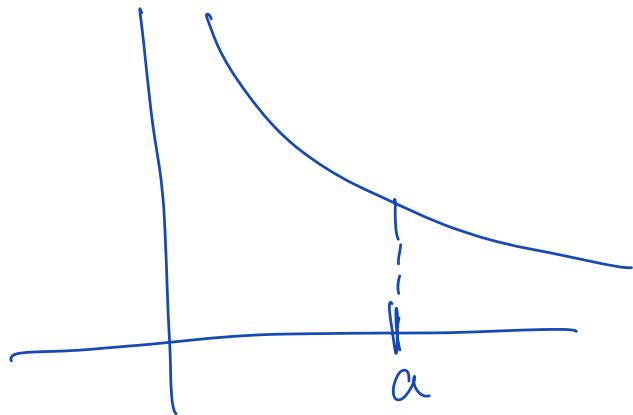


not pdf!
 $\int_{-\infty}^{+\infty} f(x) dx = 1$



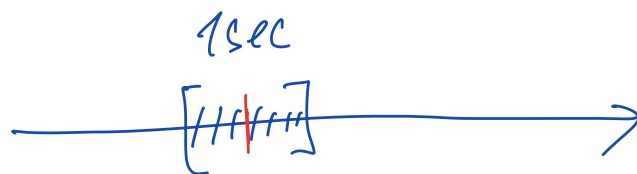
$$P\{\underline{a} < X \leq \overset{b}{b}\} = \int_{\underset{a}{a}}^{\overset{b}{b}} f(x) dx$$

$$P(X=a) = \int_a^a f(x) dx = 0$$



$$P(X=a) = P(X \in [a, a]) = \int_a^a f(x) dx = 0$$

for any number a $P(X=a)=0$



$$\int_{-\infty}^0 c e^{\alpha x} dx + \int_0^{\infty} c e^{-\alpha x} dx$$

Example

when $x = -\infty, y = +\infty$
 $x = 0, y = 0$

$$\int_{-\infty}^0 c e^{-\alpha y} d(-y) = - \int_{+\infty}^0 c e^{-\alpha y} dy = - \left(- \int_0^{+\infty} \right) = \int_0^{+\infty} c e^{-\alpha y} dy$$

$$|x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$$

The pdf of the samples of the amplitude of speech waveforms is found to decay exponentially at rate alpha

$$f(x) = c e^{-\alpha |x|}$$

Find constant c and find probability

$$P(|X| < v)$$

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

$$\int_{-\infty}^{\infty} c e^{-\alpha |x|} dx = 2c \int_0^{\infty} e^{-\alpha x} dx = 2c \left[\frac{e^{-\alpha x}}{-\alpha} \right]_0^{\infty} =$$

$$= 0 + \frac{2c}{\alpha} \cdot 1 = \frac{2c}{\alpha} = 1$$

$$\Rightarrow c = \frac{\alpha}{2}$$



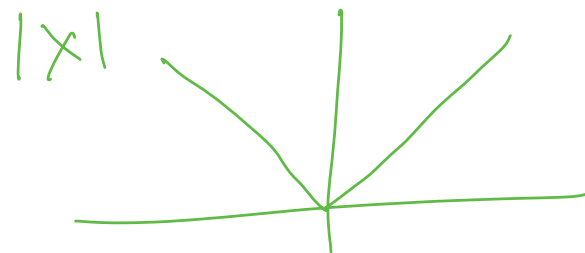
$$P(|X| < v) = \int_{-v}^v f(x) dx$$

$$\int_0^{\infty} e^{-2x} dx = -\frac{e^{-2x}}{2} \Big|_0^{\infty} = 0 - \left(-\frac{1}{2}\right) = \frac{1}{2}$$

$$\int_0^{\infty} e^{-x} dx =$$

$$\int e^x dx = e^x$$

$$\int e^{-2x} dx = -\frac{e^{-2x}}{2}$$



$$|x| = \begin{cases} x, & x > 0 \\ -x, & x < 0 \end{cases}$$

if $x = -2$, $|x| = 2$
 $-x = -(-2) = 2$

$$\left(-\frac{e^{-2x}}{2}\right) \Big|_{-\infty}^{+\infty} = e^{-2x} \Big|_{-\infty}^0 + \int_0^{+\infty} = 2 \int_0^{\infty}$$

Example speech amplitude

- Use normalization condition:

$$\int e^x dx$$

$$\int e^{cx} dx$$

$$\int x^n dx$$

$$1 = \int_{-\infty}^{+\infty} c e^{-\alpha|x|} dx = 2c \int_0^{\infty} e^{-\alpha x} dx =$$

$$= 2c \left. \frac{e^{-\alpha x}}{-\alpha} \right|_0^{\infty} = 2c \left(0 + \frac{1}{\alpha} \right) = \frac{2c}{\alpha}$$

$$\Rightarrow c = \frac{\alpha}{2}$$

- Find probability

$$P\{|X| < v\} = \frac{\alpha}{2} \int_{-v}^v e^{-\alpha|x|} dx = 2 \cdot \frac{\alpha}{2} \int_0^v e^{-\alpha x} dx =$$

$$= \alpha \left. \frac{e^{-\alpha x}}{-\alpha} \right|_0^v = \underline{\underline{[1 - e^{-\alpha v}]}}$$

Cumulative distribution function

- The **cdf** of a r.v. X is defined for any real number x as the probability of the event $\{X \leq x\}$

$$\underline{F}(x) = P(X \leq x)$$

$$P(X \in (-\infty, x])$$

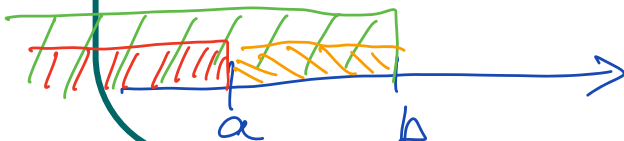
- F is a function of x .
- All probability questions about X can be answered in terms of its distribution function
- Example: how to compute $P(a < X \leq b)$?

$$P(X \leq b) = F(b)$$

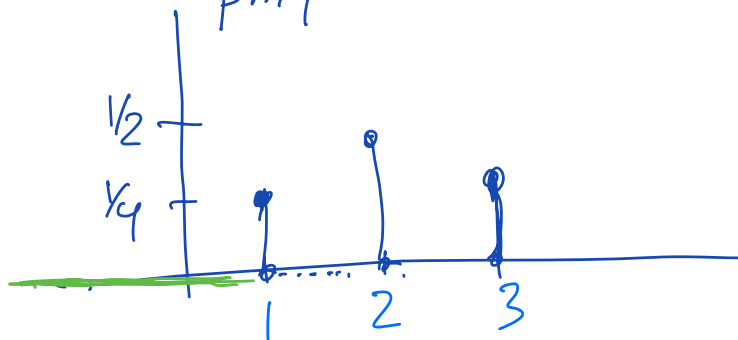
$$P(\{X \leq b\}) = P(\{X \leq a\} \cup \{a < X \leq b\})$$

$$P\{X \leq b\} = P\{X \leq a\} + P\{a < X \leq b\}$$

$$\underline{P\{a < X \leq b\} = F(b) - F(a)}$$



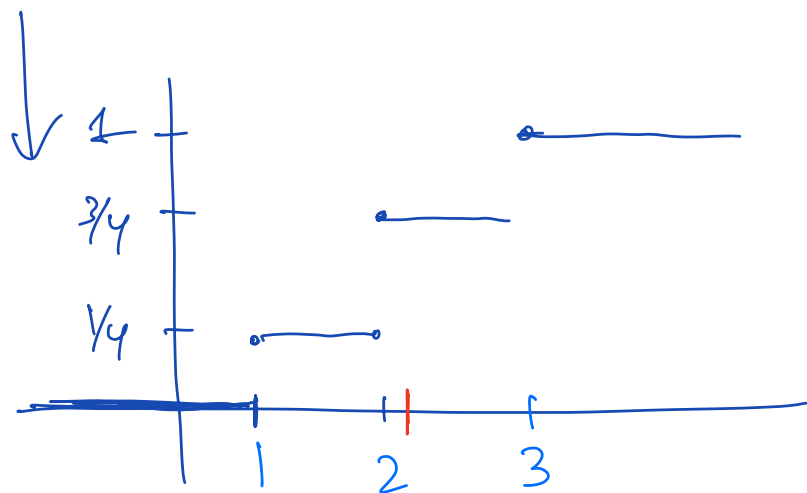
Discrete case
pmf



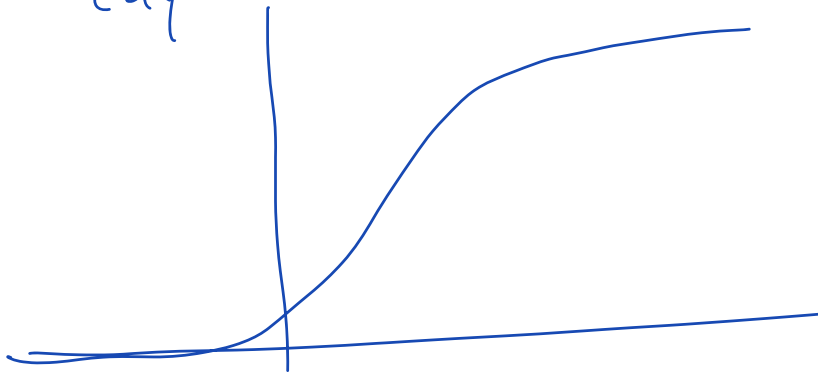
$$F(2) = P(X \leq 2) = P(X=1) + P(X=2) = \frac{3}{4}$$

pmf

cdf



Continuous case
cdf



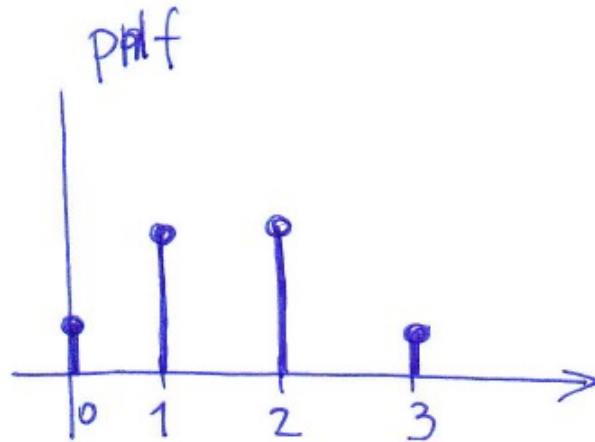
Cdf for discrete r.v.

pmf $\rightarrow$ *cdf*

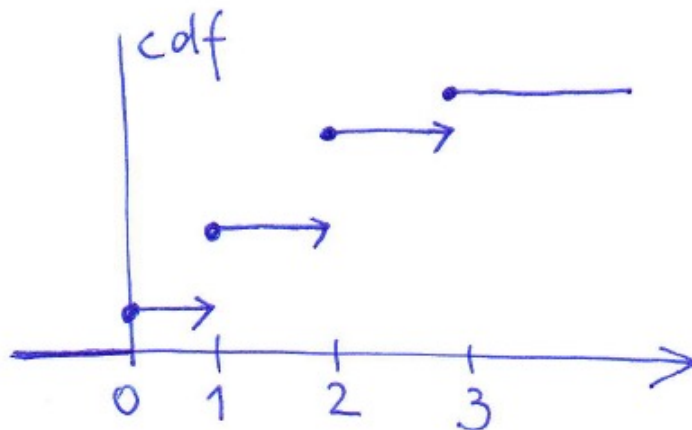
- The **cdf** is a step-function and can be expressed as

$$F(a) = \sum_{\text{all } x \leq a} p(x)$$

Example coin flipping



$$\sum_i p(i) = 1$$



$$F(a) = \sum_{\text{all } x \leq a} p(x)$$

Cdf for continuous r.v

pdf $\xrightarrow{s}$ cdf

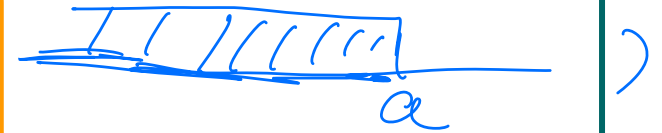
- The relationship between cdf and pdf is expressed by

$$F(a) = P(X \in (-\infty, a]) = \int_{-\infty}^a f(x) dx$$

$X \in (-\infty, a]$

cdf $\xrightarrow{\text{differentiate}}$ pdf :

$$\frac{d}{da} F(a) = f(a)$$



$$\int_{-\infty}^a f(x) dx = F(a)$$

discrete case $\xrightarrow{\sum}$ cont. case $\int$

$$F(x) = P(X \in (-\infty, x])$$

Properties of cdf

1. $0 \leq F(x) \leq 1$
 $P(X \in (-\infty, -\infty)) = 0$
2. $\lim_{x \rightarrow \infty} F(x) = 1$

not possible

3. $\lim_{x \rightarrow -\infty} F(x) = 0$ $x = -\infty$

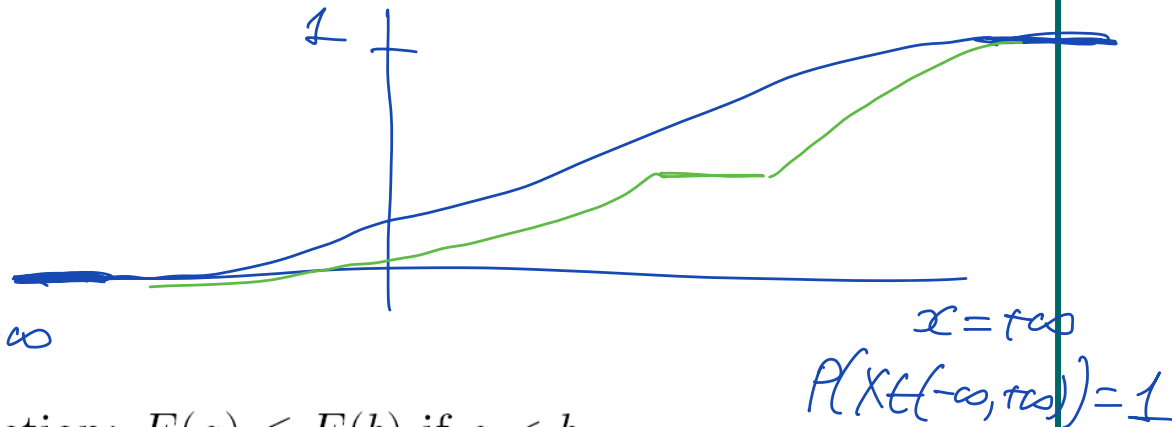
4. $F(x)$ is a nondecreasing function: $F(a) \leq F(b)$ if $a < b$

5. $F(x)$ is a continuous from the right:

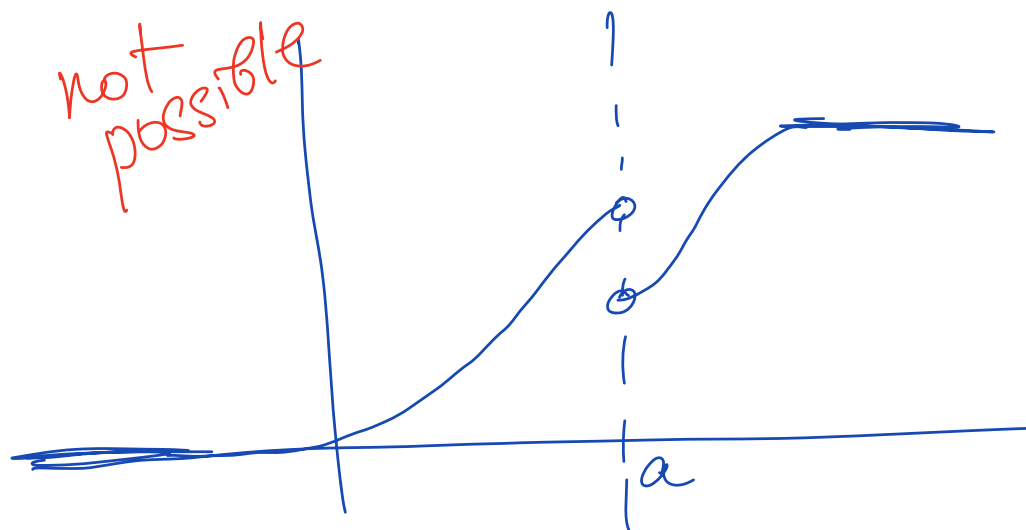
$$F(b) = \lim_{h \rightarrow 0} F(b+h) = F(b+)$$

6. Probability that a r.v. X takes on a specific value b is equal to the jump (step) of cdf at the point b :

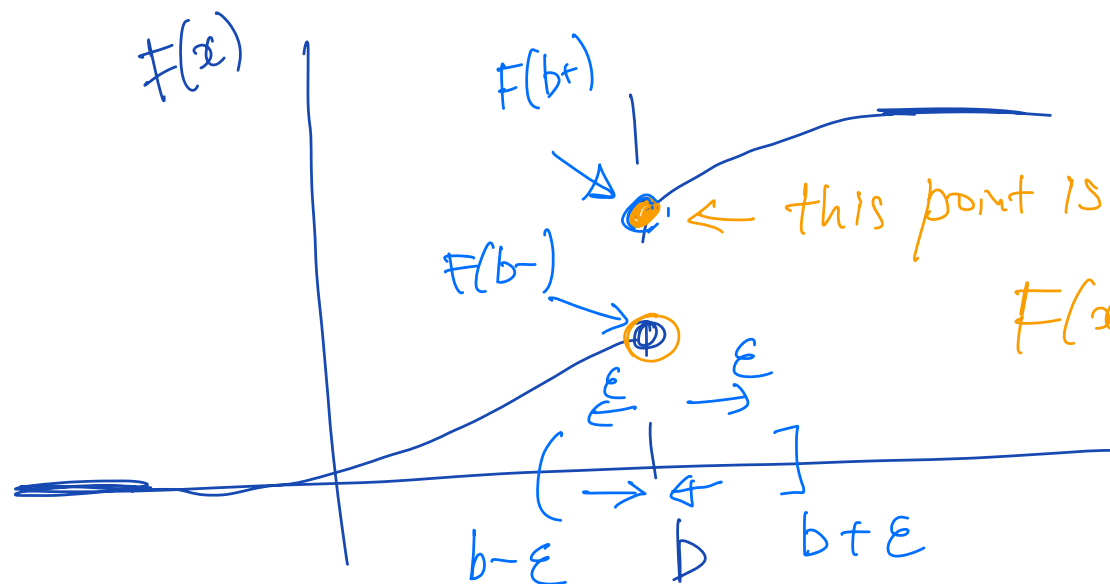
$$P(X = b) = F(b+) - F(b-)$$



not possible



$$F(x) = P(X \leq x)$$



← this point is included

$F(x)$ is continuous from the right

$$P(X=b)$$

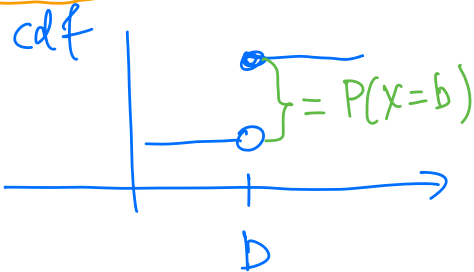
$$\varepsilon \rightarrow 0$$

$$P(X \in (b-\varepsilon, b+\varepsilon]) = F(b+\varepsilon) - F(b-\varepsilon)$$

$$P(X \in (a, b]) = F(b) - F(a) \quad \downarrow \varepsilon \rightarrow 0$$

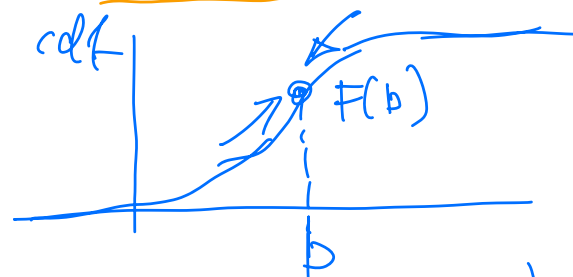
$$P(X=b) = F(b+) - F(b-)$$

Discrete r.v.



$$P(X=b) = F(b+) - F(b-) = \text{"jump"}$$

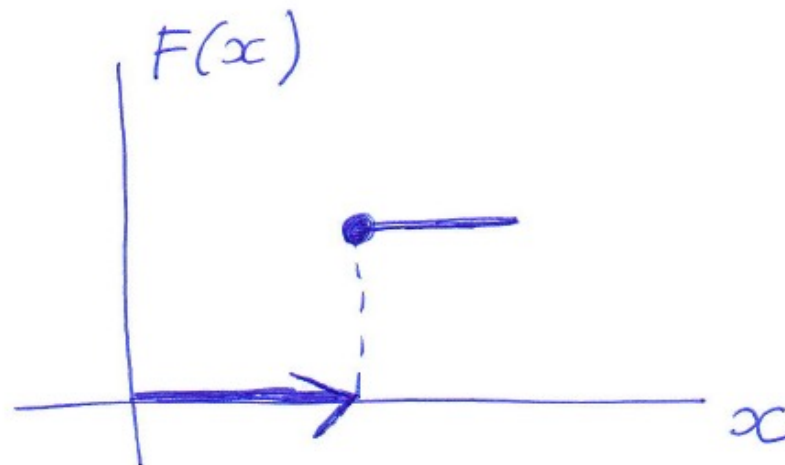
Cont. r.v.



$$P(X=b) = F(b) - F(b) = \underline{0}$$

Property no 5

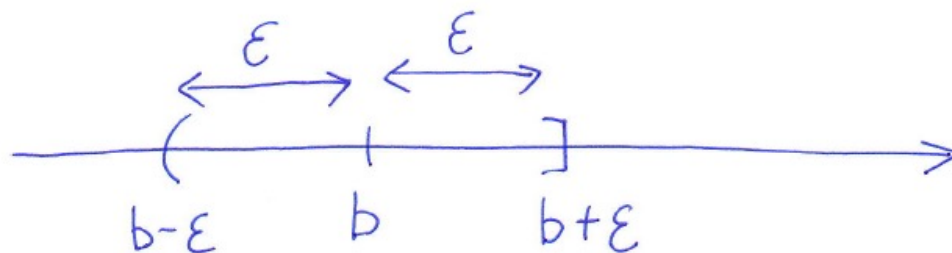
Continuous from the right:



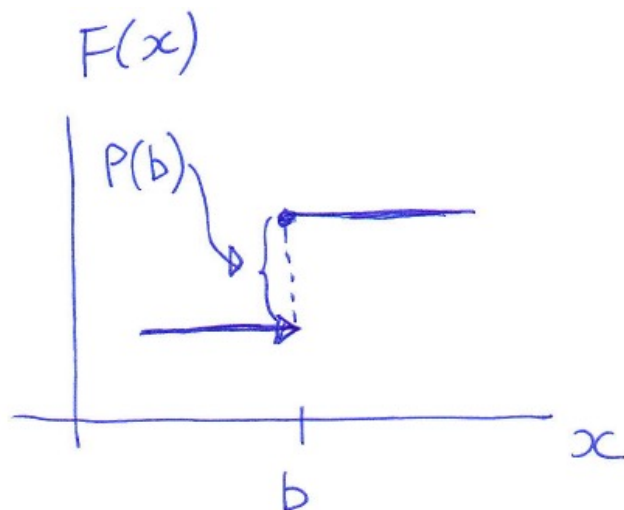
Property no 6

We know $P\{a < X \leq b\} = F(b) - F(a)$

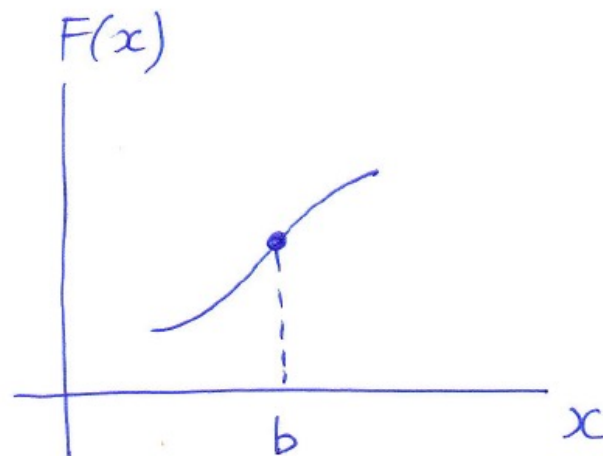
$$\begin{aligned} P\{X=b\} &= \lim_{\varepsilon \rightarrow 0} P\{b-\varepsilon < X \leq b+\varepsilon\} = \\ &= \lim_{\varepsilon \rightarrow 0} [F(b+\varepsilon) - F(b-\varepsilon)] = F(b+) - F(b-) \end{aligned}$$



Property no 6



discrete r.v.
 $P(b) = \text{"jump"}$

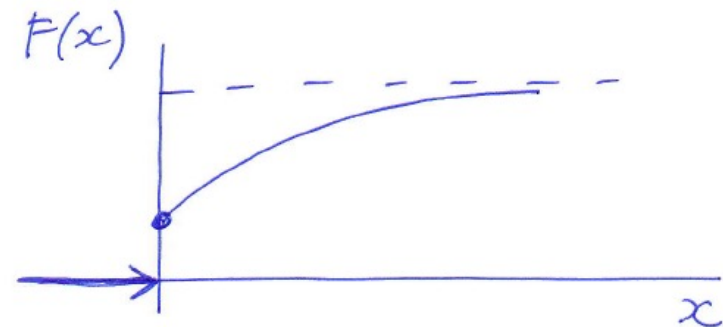
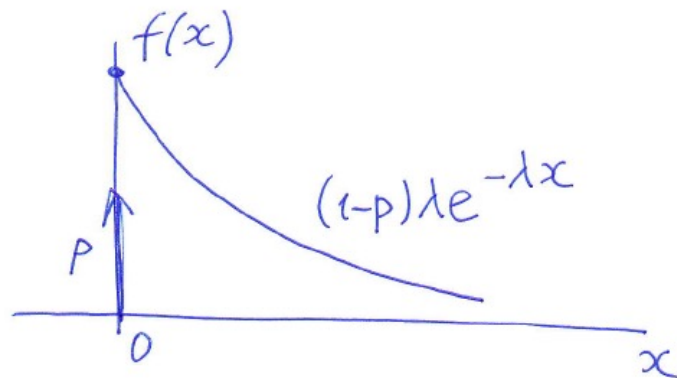


continuous r.v.
 $P(b) = 0$

Example of a mixed r.v.

- The delay (= waiting time in a queue) for a packet transmission is zero if the queue is empty, and if the queue is not empty, the delay is an exponentially distributed r.v. with cdf $F(x) = 1 - e^{-\lambda x}$
- The probability that the queue is empty is p and busy $1-p$.
- Cdf of the delay X :

$$\begin{aligned}
 F(x) &= P(X \leq x) = P(X \leq x | \text{idle}) \cdot p + P(X \leq x | \text{busy}) \cdot (1-p) \\
 &= p + (1-p)(1 - e^{-\lambda x})
 \end{aligned}$$



Types of r.v.

Discrete r.v.	Continuous r.v.	Mixed type
Pmf $p(x)$	Pdf $f(x)$	
$P\{a < X < b\} = \sum_{x_i \in (a,b)} p(x_i)$	$P(a \leq X \leq b) = \int_a^b f(x) dx$	

Multiple r.v.

- So far, we were speaking about calculation of probabilities of events involving a single r.v. in isolation. Now we will look at the techniques for probability calculations of events that involve the joint behavior of two or more r.v.
- Example: height, weight and age of a person from a group

joint
pmf /
pdf



joint
cdf

Joint cdf

$$F(X, Y, Z) = P(X \leq x, Y \leq y, Z \leq z)$$

- To specify the relationship between two r.v., we define the **joint cumulative probability distribution** function of X and Y

F_{XY}

$$F(X, Y) = P(X \leq x, Y \leq y)$$

- A knowledge of the joint cdf enables us to calculate the distribution function of r.v. X:

F_X

$$\underline{F_X(x) = P(X \leq x)} = P(X \leq x, \underline{Y < \infty}) = \underline{F(x, \infty)}_{F_{XY}}$$

Joint pmf for discrete r.vs.

- If X and Y are discrete r.vs., we define **joint probability mass function**:

$$\underline{p(x_i, y_i)} = P(\underline{X = x_i}, \underline{Y = y_i})$$

- The **individual mass functions** are easily obtained from the joint pmf:

$$\{X = x_i\} = \bigcup_j \{X = x_i, Y = y_j\}$$

joint pmf $\rightarrow$ individual pmf

$$P\{X = x_i\} = P \bigcup_j \{X = x_i, Y = y_j\} = \sum_j P\{X = x_i, \underbrace{Y = y_j}_{\text{circled}}\} = \sum_j p(x_i, y_j)$$

- The joint probabilities can be presented in tabular form. Because the individual probabilities appear in the margin of the table, they are often called **marginal probabilities**.

X, Y

$Y \backslash X$	1	2	3	
0	$\frac{1}{8}$	$\frac{1}{3}$	$\frac{1}{8}$	$\frac{7}{12}$
1	$\frac{1}{8}$	$\frac{1}{6}$	$\frac{1}{8}$	$\frac{5}{12}$
	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{4}$	

marginal pmf for Y

$$P(X=1, Y=0) = \frac{1}{8}$$

$$P(X=2, Y=0) = \frac{1}{3}$$

$\vdots$

$$P(X=1) =$$

$$P(X=1, Y=0) + P(X=1, Y=1)$$

marginal pmf for X
(= individual)

Joint pdf for continuous r.vs.

- We say that X and Y are jointly continuous, if there exist a function $f(x,y)$ defined for all real x and y , having the property that for every set C in the 2-dimensional plane

$$P\{(X,Y) \in C\} = \int \int_{(x,y) \in C} \underline{f(x,y) dx dy}$$

- $f(x,y)$ is called joint probability density function

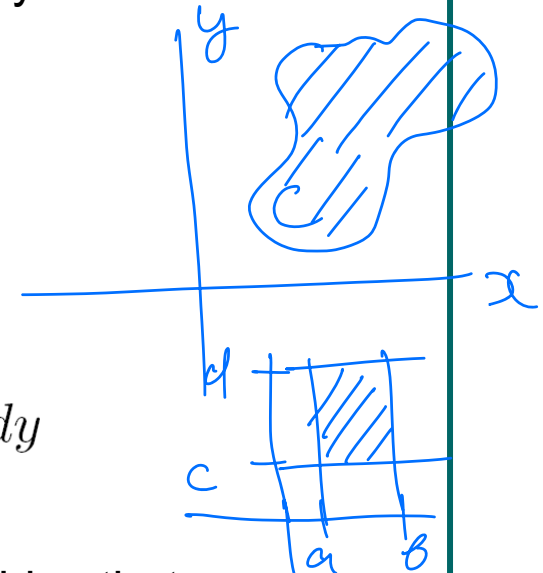
$$P\{X \in [a, b], Y \in [c, d]\} = \int_c^d \int_a^b \underline{f(x,y) dx dy}$$

- The marginal pdfs are obtained by integrating out the variables that are not of interest.

$$f_X = \int_{-\infty}^{\infty} f(x,y) dy$$

$$f_Y = \int_{-\infty}^{\infty} f(x,y) dx$$

$$f_X(x) = \int_{-\infty}^{\infty} f(x,y) dy$$



Question

- As we have seen, specifying the joint probability mass function or probability density function determines the individual distribution functions.
- Is reverse true? If I know individual mass functions, can I determine the joint mass function?

joint distribution

- pmf/pdf

- cdf



marginal
distributions



not possible
(unless extra inform.)

Example

X, Y - indep.

I

$Y \backslash X$	0	1	
0	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{2}$
1	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{2}$
	$\frac{1}{2}$	$\frac{1}{2}$	

II X, Y - not independent

$Y \backslash X$	0	1	
0	$\frac{1}{2}$	0	$\frac{1}{2}$
1	0	$\frac{1}{2}$	$\frac{1}{2}$
	$\frac{1}{2}$	$\frac{1}{2}$	

$$P(X=i, Y=j) = P(X=i) \cdot P(Y=j) \leftarrow \text{def. of independence}$$

X - transmit $\{0, 1\}$

Y - receive $\{0, 1\}$

Example: transmission system

$T_F \backslash R$		0	1	
0	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{2}$	
1	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{2}$	
	$\frac{1}{2}$	$\frac{1}{2}$		

$T_F \backslash R$		0	1	
0	$\frac{1}{2}$	0	$\frac{1}{2}$	
1	0	$\frac{1}{2}$	$\frac{1}{2}$	
	$\frac{1}{2}$	$\frac{1}{2}$		

Independent r.v.

- **Definition.** X and Y are **independent**, if for any two sets of real numbers A and B

→ $P(X \in A, Y \in B) = P(X \in A)P(Y \in B)$

- In terms of joint cumulative distribution function:

→ $F(a, b) = F_X(a)F_Y(b)$

- In terms of pmf (discrete r.v.) and pdf (continuous r.v.)

$$p(x, y) = p_X(x)p_Y(y)$$

$$f(x, y) = f_X(x)f_Y(y)$$

- Basically, X and Y are independent, if knowing the value of one does not change the distribution of another

Multiple r.v.

- Let $X_1, \dots, X_n$ be the **jointly distributed random variables**.
- Definitions for pmf, pdf and cdf can be generalized to a case of n r.v.s. For example,
- the joint cumulative distribution function is defined as

$$F(x_1, \dots, x_n) = P(X_1 \leq x_1, \dots, X_n \leq x_n)$$

- **Example.** A computer system receives messages over three communication lines. Let X_i be the number of messages received on line i in one hour. The joint pmf is given by

$$p(x_1, x_2, x_3) = (1 - a_1)(1 - a_2)(1 - a_3)a_1^{x_1}a_2^{x_2}a_3^{x_3}$$

Find individual pmfs

Example 3 communication lines

$$\begin{aligned} p_{X_3}(x_3) &= \sum_{i=1}^{\infty} \sum_{k=1}^{\infty} (1-a_1)(1-a_2)(1-a_3) a_1^i a_2^k a_3^{x_3} = \\ &= (1-a_1)(1-a_2)(1-a_3) a_3^{x_3} \underbrace{\sum_{i=1}^{\infty} a_1^i}_{\frac{1}{1-a_1}} \underbrace{\sum_{k=1}^{\infty} a_2^k}_{\frac{1}{1-a_2}} = \\ &= (1-a_3) a_3^{x_3} \end{aligned}$$

- Q: are lines independent?

Conditional distributions

- The relationship between two random variables can often be clarified by consideration of the conditional distribution of one given the value of the other.
- The **conditional pmf** of X given that $Y=y$ is defined by

$$p_{X|Y}(x|y) = \frac{p(x, y)}{p_Y(y)}$$

- If X and Y have a joint pdf, then the **conditional pdf** of X given that $Y=y$ is defined as

$$f_{X|Y}(x|y) = \frac{f(x, y)}{f_Y(y)}$$

Example conditional distributions

- Joint pmf of X and Y is given:

		X		
		0	1	
Y	0	0.4	0.1	0.5
	1	0.2	0.3	0.5
		0.6	0.4	

- Calculate the conditional pmf of X given that Y=1:

$$P\{X=0|Y=1\} = \frac{P(0,1)}{P\{Y=1\}} = \frac{0.2}{0.5} = 0.4$$

$$P\{X=1|Y=1\} = \frac{P(1,1)}{P\{Y=1\}} = \frac{0.3}{0.5} = 0.6$$